

Amit Kumar Mohanty

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SUMMARY

AI Engineer with ~1 year of experience designing, building, and deploying intelligent systems powered by LLMs, Agentic AI, and Generative AI. Recognized as "Rookie of the Year 2025" at Keywords Studios for exceeding all KPIs. Skilled in Python, LangChain/LangGraph, RAG, fine-tuning (LoRA/QLoRA), and serving AI applications at scale with FastAPI and cloud platforms (AWS, GCP Vertex AI), with a peer-reviewed publication in computer vision.

EDUCATION

K.R. Mangalam University

July 2021 – July 2025

B.Tech in Computer Science and Engineering (Specialization in AI/ML), Gurgaon | Cumulative GPA: 8.5/10

Relevant Coursework: Data Structures & Algorithms, NLP, Computer Vision, DBMS, Machine Learning, Deep Learning

EXPERIENCE

Research Associate – AI, Keywords Studios India – Gurgaon, Haryana

May 2025 – Present

- Trained and optimized enterprise-grade Agentic AI models to autonomously execute complex multi-step workflows (flight booking, e-commerce ordering, web data retrieval), improving task-completion reliability.
- Built custom data pipelines and delivered 85% of self-generated training datasets directly to production as the first member of a global AI team, accelerating model readiness.
- Awarded "Rookie of the Year 2025" for surpassing all KPIs through advanced model training and cross-functional collaboration.

Artificial Intelligence Intern, Infosys – Remote

November 2024 – February 2025

- Engineered an AI call-center system with LangChain and LangSmith, boosting response accuracy by 35% and elevating customer-AI interaction quality.
- Developed and deployed LLM features that reduced system latency by 25%, streamlining the AI deployment pipeline.
- Collaborated with 12 engineers to integrate end-to-end AI workflows and was selected to present at the Best Project by Indian Interns initiative.

Data Analytics & ML Intern, QL2 Software – Gurgaon, Haryana

June 2024 – August 2024

- Built predictive ML models for competitive pricing intelligence and performed large-scale data extraction and analysis to deliver business insights.
- Automated data collection and feature pipelines that fed downstream models with clean, structured inputs.

PROJECTS

Dynamic Screen Companion

June 2025 – Present

- Built a real-time multi-modal AI assistant integrating the Gemini API to analyze live screen activity and deliver continuous voice feedback and interactive support.
- Engineered a high-performance FastAPI model-serving backend handling concurrent inference streams, reducing manual task time by 40%.
- Designed the end-to-end multi-modal architecture from scratch and optimized OCR pipelines to maintain 95%+ extraction accuracy across varied layouts.

Diabetic Optiscan

January 2025 – June 2025

- Led integration of a Vision Transformer (ViT-B/16) on the APTOS dataset, optimizing self-attention to capture complex spatial patterns in retinal images.
- Designed a custom wavelet modification technique that boosted detection precision by 50% over legacy methods.
- Co-authored a peer-reviewed research paper detailing the system architecture and performance breakthroughs.

Currency Detection App for Visually Impaired

July 2024 – December 2024

- Trained an image-classification model on 200,000+ banknote images achieving 96% accuracy, deployed in a mobile app with voice feedback to help prevent fraud.
- Conducted user interviews with visually impaired individuals to refine real-world accessibility features.

SKILLS

Languages & Frameworks: Python, C++, SQL, FastAPI, TensorFlow, PyTorch, LangChain, LangGraph, Hugging Face, OpenCV, Scikit-learn

Generative & Agentic AI: LLMs, Agentic AI, RAG, Fine-tuning (LoRA, QLoRA), Prompt Engineering, Transformers, BERT, Multi-modal AI, Gemini/OpenAI APIs, NLP, Vector Databases (FAISS/Chroma)

AI Systems & Deployment: Model Serving, API Development, Docker, CI/CD, MLflow, Model Evaluation & Optimization, Data Pipelines, Git

Data & Cloud: AWS, GCP (Vertex AI), MySQL, MongoDB, Pandas, NumPy, Power BI

CERTIFICATIONS

Ducat India – Data Science Professional Training

April 2024 – April 2025

Professional training in Data Science: Python, statistics, machine learning, data analysis, and predictive modeling.

AI/ML for Geodata Analysis

Offered by ISRO (Indian Space Research Organisation) – applying AI/ML to geospatial data analysis.